

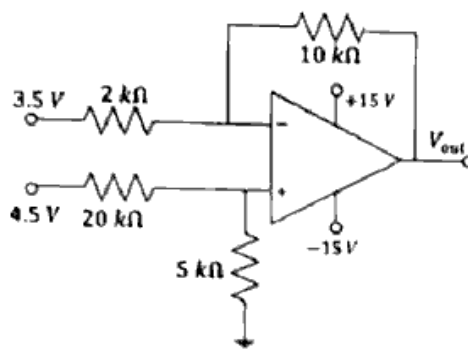
Instructions:-

- (i) The marks are indicated in the right-hand margin.
- (ii) There are NINE questions in this paper.
- (iii) Attempt FIVE questions in all.
- (iv) Question No. 1 is compulsory.

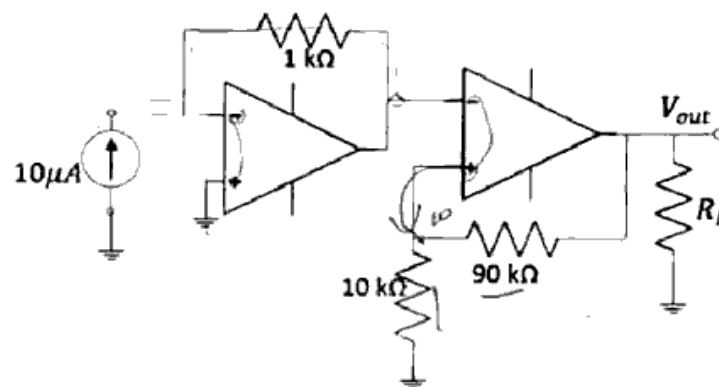
Q.1 Choose the correct option / answer the following (Any seven questions only):-**[2 x 7 = 14]**

- (a) A Zener diode rated 5.1V is used with a 220 Ω resistor. If input is 12V, what is the Zener current?
 - (i) 31.4 mA
 - (ii) 50 mA
 - (iii) 20 mA
 - (iv) 10 mA
- (b) At room temperature, a transistor has a collector current of 5 mA and $V_{BE} = 0.7V$. Find g_m .
 - (i) 7.14 S
 - (ii) 194 mS
 - (iii) 5 S
 - (iv) 3.5 S
- (c) Which region does a MOSFET operate in for amplification?
 - (i) Cutoff
 - (ii) Triode
 - (iii) Saturation
 - (iv) Ohmic
- (d) The trans-conductance (g_m) of a MOSFET is a measure of:
 - (i) Voltage gain
 - (ii) Input impedance
 - (iii) Output conductance
 - (iv) Change in drain current per unit change in gate voltage
- (e) Which parameter limits the maximum frequency response of an op-amp?
 - (i) Slew rate
 - (ii) Offset voltage
 - (iii) Input bias current
 - (iv) Common-mode gain
- (f) An inverting amplifier has $R_f = 100\text{ k}\Omega$ and $R_{in} = 10\text{ k}\Omega$. What is the voltage gain?
 - (i) -100
 - (ii) -10
 - (iii) 10
 - (iv) 1
- (g) Which op-amp circuit converts a sinusoidal input to a square wave output?
 - (i) Peak detector
 - (ii) Comparator
 - (iii) Schmitt trigger
 - (iv) Integrator
- (h) A MOSFET has $I_D = 2\text{ mA}$ and $V_{GS} = 2.5$. If $V_{th} = 1\text{ V}$ and $k_n = 1\text{ mA/V}^2$, what is the value of I_D .
 - (i) 1 mA
 - (ii) 2 mA
 - (iii) 3.5 mA
 - (iv) 4 mA
- (i) Which component is used to maintain a constant voltage across the load?
 - (i) P-N junction diode
 - (ii) Zener diode
 - (iii) Schottky diode
 - (iv) Tunnel diode
- (j) Which configuration of BJT offers the highest current gain?
 - (i) Common Base
 - (ii) Common Collector
 - (iii) Common Emitter
 - (iv) Emitter Follower

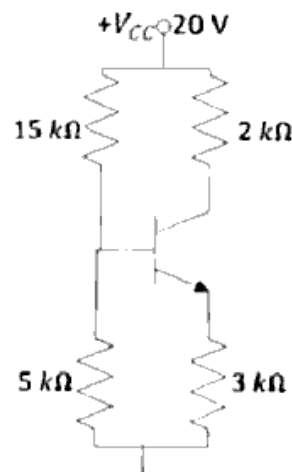
- Q.2** (a) Explain the working of a differential amplifier using BJTs. Derive expressions for differential and common-mode gain. [7]
- (b) Assume the op-amp is ideal, find the output voltage V_{out} . [7]



- Q.3** (a) Design and analyze a Wein Bridge Oscillator using op-amp. Derive the expression for frequency of oscillation. [7]
- (b) Explain the operation of a Zener diode as a voltage regulator. Design a circuit to provide 5V regulated output. [7]
- Q.4** (a) Explain the operation of a hysteretic comparator using op-amp. Draw and analyze the transfer characteristics. [7]
- (b) Assume the op-amp is ideal, find the output voltage V_{out} . [7]



- Q.5** (a) Design and analyze a clamping circuit. Explain how it shifts the DC level of an input waveform. [7]
- (b) Explain the effect of channel length modulation on the performance of a MOSFET amplifier. [7]
- Q.6** (a) Design a triangular wave generator using op-amps. [7]
- (b) For the circuit shown below, draw D.C. load line. Assume β is very large. [7]



- Q.7 (a) Design a first-order low-pass butter worth filter circuit using op-amp and derive the expression of high cut off frequency and transfer function of the filter. [7]
(b) Explain the role of op-amp in waveform shaping circuits with examples. [7]
- Q.8 (a) Derive the expression for ripple voltage and peak inverse voltage in a full-wave rectifier. [7]
(b) Explain the various biasing techniques for a BJT amplifier. Suggest which one is better and why? [7]
- Q.9 Write short notes on *any two* of the following:- [7 x 2 = 14]
(a) Current mirror using BJT
(b) Trans-conductance of MOSFET
(c) Junction break-down in Zener diodes
(d) high-frequency equivalent circuit of BJT
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